## What is Pandas?

Python library used for working with structured data sets (tables). It has functions for analyzing, cleaning, exploring, and manipulating data. Similar to data tables in SQL and Excel.

```
# import pandas
import pandas as pd
import numpy as np
```

```
pd.__version__ # check installed pandas version
```

```
'2.2.3'
```

## What is Dataframe?

It's a structured data constructed with rows and columns, similar to a SQL tables or Excel tables

```
#Create Dataframe
df = pd.DataFrame([20,22,323], columns=['Marks'])
print(df)
```

```
   Marks
0     20
1     22
2    323
```

```
# using Dictionary of Lists
data = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi','abc','xyz','wqr'],
    'S_Age': [16,17,18,19,20,21,23],
    'Marks': [90, 70, 55, 56,30,58,54]
}
```

```
print(type(data))
```

```
<class 'dict'>
```

```
df = pd.DataFrame(data)
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |
| **1** | Reena | 17 | 70 |
| **2** | Shyam | 18 | 55 |
| **3** | Ravi | 19 | 56 |
| **4** | abc | 20 | 30 |
| **5** | xyz | 21 | 58 |
| **6** | wqr | 23 | 54 |

```
df.head(2)
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |
| **1** | Reena | 17 | 70 |

```
df.tail(2)
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **5** | xyz | 21 | 58 |
| **6** | wqr | 23 | 54 |

```
df.shape
```

```
(7, 3)
```

```
#column name
df.columns
```

```
Index(['S_Name', 'S_Age', 'Marks'], dtype='object')
```

```
df.to_csv('student.csv',index=False)
```

# inplace=True : "Make the change directly in the original DataFrame."

```
# rename column name
df.rename(columns={'Marks': 'End_term_marks'}, inplace=True)
```

```
df
```

| | S_Name | S_Age | End_term_marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |
| **1** | Reena | 17 | 70 |
| **2** | Shyam | 18 | 55 |
| **3** | Ravi | 19 | 56 |
| **4** | abc | 20 | 30 |
| **5** | xyz | 21 | 58 |
| **6** | wqr | 23 | 54 |

```
df.columns
```

```
Index(['S_Name', 'S_Age', 'End_term_marks'], dtype='object')
```

```
df.rename(columns={
    'S_Name': 'StudentName',
    'S_Age': 'StudentAge',
    }, inplace=True)
```

```
df
```

| | StudentName | StudentAge | End_term_marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |
| **1** | Reena | 17 | 70 |
| **2** | Shyam | 18 | 55 |
| **3** | Ravi | 19 | 56 |
| **4** | abc | 20 | 30 |
| **5** | xyz | 21 | 58 |
| **6** | wqr | 23 | 54 |

```
# info method prints information about the DataFrame
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column          Non-Null Count  Dtype 
---  ------          --------------  ----- 
 0   StudentName     7 non-null      object
 1   StudentAge      7 non-null      int64 
 2   End_term_marks  7 non-null      int64 
dtypes: int64(2), object(1)
memory usage: 300.0+ bytes
```

```
# describe method generates descriptive statistics of DataFrame,
#only for numerical-value columns
df.describe()
```

| | StudentAge | End_term_marks |
|---|---|---|
| **count** | 7.000000 | 7.000000 |
| **mean** | 19.142857 | 59.000000 |
| **std** | 2.410295 | 18.138357 |
| **min** | 16.000000 | 30.000000 |
| **25%** | 17.500000 | 54.500000 |
| **50%** | 19.000000 | 56.000000 |
| **75%** | 20.500000 | 64.000000 |
| **max** | 23.000000 | 90.000000 |

# Rows & Columns - Selection

```
# select multiple columns
df['StudentName']
```

| | StudentName |
|---|---|
| **0** | Amit |
| **1** | Reena |
| **2** | Shyam |
| **3** | Ravi |
| **4** | abc |
| **5** | xyz |
| **6** | wqr |

**dtype:** object

```
# select multiple columns
df[['StudentName', 'End_term_marks']]
```

| | StudentName | End_term_marks |
|---|---|---|
| **0** | Amit | 90 |
| **1** | Reena | 70 |
| **2** | Shyam | 55 |
| **3** | Ravi | 56 |
| **4** | abc | 30 |
| **5** | xyz | 58 |
| **6** | wqr | 54 |

```
df.loc[df.StudentName=='Amit']
```

| | StudentName | StudentAge | End_term_marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |

```
df.loc[(df.StudentName=='Amit') & (df.End_term_marks>=90)]
```

| | StudentName | StudentAge | End_term_marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |

```
df.shape
```

```
(7, 3)
```

## Filter Dataframe - Filtering by column values

```
df_age_filter = df[df['StudentAge'] >= 20] # filter and store dataframe in a ne
```

```
df_age_filter
```

| | StudentName | StudentAge | End_term_marks |
|---|---|---|---|
| **4** | abc | 20 | 30 |
| **5** | xyz | 21 | 58 |
| **6** | wqr | 23 | 54 |

## Rows and Columns - Operation (Add, update, delete)

```
df
```

| | StudentName | StudentAge | End_term_marks |
|---|---|---|---|
| **0** | Amit | 16 | 90 |
| **1** | Reena | 17 | 70 |
| **2** | Shyam | 18 | 55 |
| **3** | Ravi | 19 | 56 |
| **4** | abc | 20 | 30 |
| **5** | xyz | 21 | 58 |
| **6** | wqr | 23 | 54 |

```
# Create a new column
df['Department'] = ['B.Tech(EC)', 'B.Tech(EE)', 'B.Tech(CS)', 'B.Tech(AIML)','B
df
```

| | StudentName | StudentAge | End_term_marks | Department |
|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) |
| **1** | Reena | 17 | 70 | B.Tech(EE) |
| **2** | Shyam | 18 | 55 | B.Tech(CS) |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) |
| **4** | abc | 20 | 30 | B.Tech(EC) |
| **5** | xyz | 21 | 58 | B.Tech(CS) |
| **6** | wqr | 23 | 54 | B.Tech(EE) |

```
# Add new columns using existing column/s
df['Grade'] = np.where(df['End_term_marks'] >= 90, 'A',
                np.where(df['End_term_marks'] >= 80, 'B',
                np.where(df['End_term_marks'] >= 70, 'C',
                np.where(df['End_term_marks'] >= 60, 'D', 'F'))))

df
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **1** | Reena | 17 | 70 | B.Tech(EE) | C |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 30 | B.Tech(EC) | F |
| **5** | xyz | 21 | 58 | B.Tech(CS) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |

```
# Add new row - at the end of dataframe
df.loc[len(df)] = ['Reema', 21, 90, 'IT', 2000]
df
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **1** | Reena | 17 | 70 | B.Tech(EE) | C |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 30 | B.Tech(EC) | F |
| **5** | xyz | 21 | 58 | B.Tech(CS) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |
| **7** | Reema | 21 | 90 | IT | 2000 |

```
# update value in dataframe using index-name
df.loc[4, 'End_term_marks'] = 55
df
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **1** | Reena | 17 | 70 | B.Tech(EE) | C |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 55 | B.Tech(EC) | F |
| **5** | xyz | 21 | 58 | B.Tech(CS) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |
| **7** | Reema | 21 | 90 | IT | 2000 |

```
# update value in dataframe using column-value
df.loc[df.StudentName=='Reena','End_term_marks'] = 80
df
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **1** | Reena | 17 | 80 | B.Tech(EE) | C |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 55 | B.Tech(EC) | F |
| **5** | xyz | 21 | 58 | B.Tech(CS) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |
| **7** | Reema | 21 | 90 | IT | 2000 |

## delete value - rows and columns

```
df
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **1** | Reena | 17 | 80 | B.Tech(EE) | C |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 55 | B.Tech(EC) | F |
| **5** | xyz | 21 | 58 | B.Tech(CS) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |
| **7** | Reema | 21 | 90 | IT | 2000 |

```
# delete row using column-value filter
df = df.drop(df[df.StudentName== 'xyz'].index) # delete row, axis= 0
```

```
df
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **1** | Reena | 17 | 80 | B.Tech(EE) | C |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 55 | B.Tech(EC) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |
| **7** | Reema | 21 | 90 | IT | 2000 |

```
# delete row using index-name filter
df.drop(1, axis=0)
```

| | StudentName | StudentAge | End_term_marks | Department | Grade |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | A |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | F |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | F |
| **4** | abc | 20 | 55 | B.Tech(EC) | F |
| **6** | wqr | 23 | 54 | B.Tech(EE) | F |
| **7** | Reema | 21 | 90 | IT | 2000 |

```
df.drop('Grade', axis=1, inplace=True) # delete one column
# df.drop(['Grade', 'Department'], axis=1, inplace=True) # delete multiple colu
```

```
df
```

| | StudentName | StudentAge | End_term_marks | Department |
|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) |
| **1** | Reena | 17 | 80 | B.Tech(EE) |
| **2** | Shyam | 18 | 55 | B.Tech(CS) |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) |
| **4** | abc | 20 | 55 | B.Tech(EC) |
| **6** | wqr | 23 | 54 | B.Tech(EE) |
| **7** | Reema | 21 | 90 | IT |

```
# sort values - order values in dataframe by asc or desc order
df.sort_values('End_term_marks') #ascending order, by default
```

| | StudentName | StudentAge | End_term_marks | Department |
|---|---|---|---|---|
| **6** | wqr | 23 | 54 | B.Tech(EE) |
| **2** | Shyam | 18 | 55 | B.Tech(CS) |
| **4** | abc | 20 | 55 | B.Tech(EC) |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) |
| **1** | Reena | 17 | 80 | B.Tech(EE) |
| **0** | Amit | 16 | 90 | B.Tech(EC) |
| **7** | Reema | 21 | 90 | IT |

```
# sort values in descending order
df.sort_values('End_term_marks', ascending=False)
```

| | StudentName | StudentAge | End_term_marks | Department |
|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) |
| **7** | Reema | 21 | 90 | IT |
| **1** | Reena | 17 | 80 | B.Tech(EE) |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) |
| **2** | Shyam | 18 | 55 | B.Tech(CS) |
| **4** | abc | 20 | 55 | B.Tech(EC) |
| **6** | wqr | 23 | 54 | B.Tech(EE) |

```
df['Birth Year'] = 2026 - df['StudentAge']
```

```
df
```

| | StudentName | StudentAge | End_term_marks | Department | Birth Year |
|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | 2010 |
| **1** | Reena | 17 | 80 | B.Tech(EE) | 2009 |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | 2008 |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | 2007 |
| **4** | abc | 20 | 55 | B.Tech(EC) | 2006 |
| **6** | wqr | 23 | 54 | B.Tech(EE) | 2003 |
| **7** | Reema | 21 | 90 | IT | 2005 |

```
# create a date column - date of Birth
df['DOB'] = ['2010-01-01', '2009-01-15', '2008-03-28', '2007-03-03','2003-03-03
df
```

| | StudentName | StudentAge | End_term_marks | Department | Birth Year | DOB |
|---|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | 2010 | 2010-01-01 |
| **1** | Reena | 17 | 80 | B.Tech(EE) | 2009 | 2009-01-15 |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | 2008 | 2008-03-28 |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | 2007 | 2007-03-03 |
| | | | | | | 2003-03- |

## your DOB column is currently text (string).

## Convert it to date format

```
df['DOB'] = pd.to_datetime(df['DOB'])
```

```
df
```

| | StudentName | StudentAge | End_term_marks | Department | Birth Year | DOB |
|---|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | 2010 | 2010-01-01 |
| **1** | Reena | 17 | 80 | B.Tech(EE) | 2009 | 2009-01-15 |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | 2008 | 2008-03-28 |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | 2007 | 2007-03-03 |
| | | | | | | 2003-03- |

```
df.to_csv("student.csv",index=False)
```

## Handling Missing Values

```
df
```

| | StudentName | StudentAge | End_term_marks | Department | Birth Year | DOB |
|---|---|---|---|---|---|---|
| **0** | Amit | 16 | 90 | B.Tech(EC) | 2010 | 2010-01-01 |
| **1** | Reena | 17 | 80 | B.Tech(EE) | 2009 | 2009-01-15 |
| **2** | Shyam | 18 | 55 | B.Tech(CS) | 2008 | 2008-03-28 |
| **3** | Ravi | 19 | 56 | B.Tech(AIML) | 2007 | 2007-03-03 |
| | | | | | | 2003-03- |

## Detect Missing Values

```
df.isnull() #  Detect missing values
```

| | StudentName | StudentAge | End_term_marks | Department | Birth Year | DOB |
|---|---|---|---|---|---|---|
| **0** | False | False | False | False | False | False |
| **1** | False | False | False | False | False | False |
| **2** | False | False | False | False | False | False |
| **3** | False | False | False | False | False | False |
| **4** | False | False | False | False | False | False |
| **6** | False | False | False | False | False | False |
| **7** | False | False | False | False | False | False |

## Count missing values

```
df.isnull().sum() #  Detect missing values
```

| | 0 |
|---|---|
| **StudentName** | 0 |
| **StudentAge** | 0 |
| **End_term_marks** | 0 |
| **Department** | 0 |
| **Birth Year** | 0 |
| **DOB** | 0 |

**dtype:** int64

```
Student = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi','abc','xyz','wqr','AAA','BBB','
    'S_Age': [16,17,18,np.nan,20,21,23,20,np.nan,19],
    'Marks': [90, 70, 55, 56,30,np.nan,54,99,np.nan,78]
}
```

```
df=pd.DataFrame(Student)
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | NaN |
| **6** | wqr | 23.0 | 54.0 |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | NaN |
| **9** | CCC | 19.0 | 78.0 |

```
df.loc[df.S_Name=='wqr', 'Marks'] = np.nan # adding a null value
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | NaN |
| **6** | wqr | 23.0 | NaN |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | NaN |
| **9** | CCC | 19.0 | 78.0 |

# Deletion Method

```
#Remove rows
df.dropna()
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **4** | abc | 20.0 | 30.0 |
| **7** | AAA | 20.0 | 99.0 |
| **9** | CCC | 19.0 | 78.0 |

```
#Remove columns
df.dropna(axis=1)
```

| | S_Name |
|---|---|
| **0** | Amit |
| **1** | Reena |
| **2** | Shyam |
| **3** | Ravi |
| **4** | abc |
| **5** | xyz |
| **6** | wqr |
| **7** | AAA |
| **8** | BBB |
| **9** | CCC |

# Mean Imputation

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | NaN |
| **6** | wqr | 23.0 | NaN |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | NaN |
| **9** | CCC | 19.0 | 78.0 |

```
Student = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi','abc','xyz','wqr','AAA','BBB','
    'S_Age': [16,17,18,np.nan,20,21,23,20,np.nan,19],
    'Marks': [90, 70, 55, 56,30,np.nan,54,99,np.nan,78]
}
```

```
df=pd.DataFrame(Student)
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | NaN |
| **6** | wqr | 23.0 | 54.0 |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | NaN |
| **9** | CCC | 19.0 | 78.0 |

```
df['Marks'] = df['Marks'].fillna(df['Marks'].mean())
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | 66.5 |
| **6** | wqr | 23.0 | 54.0 |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | 66.5 |
| **9** | CCC | 19.0 | 78.0 |

```
Start coding or generate with AI.
```

# Median Imputation

```
df['Marks'] = df['Marks'].fillna(df['Marks'].median())
```

```
Student = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi','abc','xyz','wqr','AAA','BBB','
    'S_Age': [16,17,18,np.nan,20,21,23,20,np.nan,19],
    'Marks': [90, 70, 55, 56,30,np.nan,54,99,np.nan,78]
}
```

```
df=pd.DataFrame(Student)
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | NaN |
| **6** | wqr | 23.0 | 54.0 |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | NaN |
| **9** | CCC | 19.0 | 78.0 |

```
df['Marks'] = df['Marks'].fillna(df['Marks'].median())
```

```
df
```

| | S_Name | S_Age | Marks |
|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 |
| **1** | Reena | 17.0 | 70.0 |
| **2** | Shyam | 18.0 | 55.0 |
| **3** | Ravi | NaN | 56.0 |
| **4** | abc | 20.0 | 30.0 |
| **5** | xyz | 21.0 | 63.0 |
| **6** | wqr | 23.0 | 54.0 |
| **7** | AAA | 20.0 | 99.0 |
| **8** | BBB | NaN | 63.0 |
| **9** | CCC | 19.0 | 78.0 |

# Mode Imputation

```
df['Gender'] = ['Male', 'Female', 'Male', 'Male', np.nan,np.nan,np.nan,np.nan,r
```

```
df['Gender'] = df['Gender'].fillna(df['Gender'].mode()[0])
```

```
df
```

| | S_Name | S_Age | Marks | Gender |
|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male |
| **1** | Reena | 17.0 | 70.0 | Female |
| **2** | Shyam | 18.0 | 55.0 | Male |
| **3** | Ravi | NaN | 56.0 | Male |
| **4** | abc | 20.0 | 30.0 | Male |
| **5** | xyz | 21.0 | 63.0 | Male |
| **6** | wqr | 23.0 | 54.0 | Male |
| **7** | AAA | 20.0 | 99.0 | Male |
| **8** | BBB | NaN | 63.0 | Male |
| **9** | CCC | 19.0 | 78.0 | Female |

# Constant Value Imputation

```
df
```

| | S_Name | S_Age | Marks | Gender |
|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male |
| **1** | Reena | 17.0 | 70.0 | Female |
| **2** | Shyam | 18.0 | 55.0 | Male |
| **3** | Ravi | NaN | 56.0 | Male |
| **4** | abc | 20.0 | 30.0 | Male |
| **5** | xyz | 21.0 | 63.0 | Male |
| **6** | wqr | 23.0 | 54.0 | Male |
| **7** | AAA | 20.0 | 99.0 | Male |
| **8** | BBB | NaN | 63.0 | Male |
| **9** | CCC | 19.0 | 78.0 | Female |

```
df['Marks'] = df['Marks'].fillna(0)
```

# Categorical

```
df['Department'] = ['CSE', 'ECE', np.nan, 'CSE', 'IT',
                    np.nan, 'ECE', 'CSE', 'IT', np.nan]

df
```

| | S_Name | S_Age | Marks | Gender | Department |
|---|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male | CSE |
| **1** | Reena | 17.0 | 70.0 | Female | ECE |
| **2** | Shyam | 18.0 | 55.0 | Male | NaN |
| **3** | Ravi | NaN | 56.0 | Male | CSE |
| **4** | abc | 20.0 | 30.0 | Male | IT |
| **5** | xyz | 21.0 | 63.0 | Male | NaN |
| **6** | wqr | 23.0 | 54.0 | Male | ECE |
| **7** | AAA | 20.0 | 99.0 | Male | CSE |
| **8** | BBB | NaN | 63.0 | Male | IT |
| **9** | CCC | 19.0 | 78.0 | Female | NaN |

```
df['Department'] = df['Department'].fillna('Unknown')
```

```
df
```

| | S_Name | S_Age | Marks | Gender | Department |
|---|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male | CSE |
| **1** | Reena | 17.0 | 70.0 | Female | ECE |
| **2** | Shyam | 18.0 | 55.0 | Male | Unknown |
| **3** | Ravi | NaN | 56.0 | Male | CSE |
| **4** | abc | 20.0 | 30.0 | Male | IT |
| **5** | xyz | 21.0 | 63.0 | Male | Unknown |
| **6** | wqr | 23.0 | 54.0 | Male | ECE |
| **7** | AAA | 20.0 | 99.0 | Male | CSE |
| **8** | BBB | NaN | 63.0 | Male | IT |
| **9** | CCC | 19.0 | 78.0 | Female | Unknown |

## Forward Fill — ffill()

```
Student = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi', 'abc', 'xyz', 'wqr', 'AAA', 'E

    'S_Age': [16, 17, 18, np.nan, 20, 21, 23, 20, np.nan, 19],

    'Marks': [90, 70, 55, 56, 30, np.nan, 54, 99, np.nan, 78],

    'Gender': ['Male', 'Female', 'Male', np.nan, 'Female',
               'Male', 'Female', 'Male', np.nan, 'Male']
}

df = pd.DataFrame(Student)

df
```

| | S_Name | S_Age | Marks | Gender |
|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male |
| **1** | Reena | 17.0 | 70.0 | Female |
| **2** | Shyam | 18.0 | 55.0 | Male |
| **3** | Ravi | NaN | 56.0 | NaN |
| **4** | abc | 20.0 | 30.0 | Female |
| **5** | xyz | 21.0 | NaN | Male |
| **6** | wqr | 23.0 | 54.0 | Female |
| **7** | AAA | 20.0 | 99.0 | Male |
| **8** | BBB | NaN | NaN | NaN |
| **9** | CCC | 19.0 | 78.0 | Male |

```
df['Marks'] = df['Marks'].ffill()
```

```
df
```

| | S_Name | S_Age | Marks | Gender |
|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male |
| **1** | Reena | 17.0 | 70.0 | Female |
| **2** | Shyam | 18.0 | 55.0 | Male |
| **3** | Ravi | NaN | 56.0 | NaN |
| **4** | abc | 20.0 | 30.0 | Female |
| **5** | xyz | 21.0 | 30.0 | Male |
| **6** | wqr | 23.0 | 54.0 | Female |
| **7** | AAA | 20.0 | 99.0 | Male |
| **8** | BBB | NaN | 99.0 | NaN |
| **9** | CCC | 19.0 | 78.0 | Male |

# Backward Fill — bfill()

```
df['S_Age'] = df['S_Age'].bfill()
```

```
df
```

| | S_Name | S_Age | Marks | Gender |
|---|---|---|---|---|
| **0** | Amit | 16.0 | 90.0 | Male |
| **1** | Reena | 17.0 | 70.0 | Female |
| **2** | Shyam | 18.0 | 55.0 | Male |
| **3** | Ravi | 20.0 | 56.0 | NaN |
| **4** | abc | 20.0 | 30.0 | Female |
| **5** | xyz | 21.0 | 30.0 | Male |
| **6** | wqr | 23.0 | 54.0 | Female |
| **7** | AAA | 20.0 | 99.0 | Male |
| **8** | BBB | 19.0 | 99.0 | NaN |
| **9** | CCC | 19.0 | 78.0 | Male |

# Duplicate Values

## Detect Duplicate Rows

```
Student = {
    'S_Name': ['Amit', 'Reena', 'Shyam', 'Ravi', 'abc', 'xyz', 'wqr', 'AAA', 'A

    'S_Age': [16, 17, 18, np.nan, 20, 21, 23, 20, 20, 19],

    'Marks': [90, 70, 55, 56, 30, np.nan, 54, 99, 99, 78],
```